

Foundational Mathematics for ML (MA2221)

Assignment (NumPy & Linear Algebra)

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Instructions:

- This tutorial sheet is **not for submission**.
- All problems must be solved using **NumPy**.
- Use **functions, loops, and conditional statements** wherever appropriate.
- Avoid using advanced routines not discussed in the notebook.
- Problems are marked with difficulty levels: Easy ([E]), Moderate ([M]), Challenging ([C]).

1. [E] Create a NumPy array x of length 10 containing equally spaced values between 0 and 1.
 - (a) Compute $x^2 + 2x + 1$ using vectorized operations.
 - (b) Repeat the computation using a `for` loop.
 - (c) Verify that both results are identical.

2. [E] Write a function `safe_normalize(x)` that:

- computes the Euclidean norm of x ,
- normalizes x only if $\|x\|_2 > 1$,
- otherwise returns x unchanged.

Test the function on vectors of different magnitudes.

3. [M] Given vectors $x, y \in \mathbb{R}^n$:
 - (a) Compute their dot product using a loop.
 - (b) Compute the dot product using `np.dot`.
 - (c) Compute the Euclidean distance between x and y .

Compare the results numerically.

4. [E] Given the matrix

$$X = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix},$$

compute the distance between the first row and every other row using a loop.

5. [M] Let $v \in \mathbb{R}^n$ be a fixed vector and $X \in \mathbb{R}^{m \times n}$.
 - (a) Compute the cosine similarity between v and each row of X .
 - (b) Print only those rows whose similarity exceeds 0.8.

Use a loop and an `if` condition.

6. [M] Given vectors $x, y \in \mathbb{R}^n$:
 - (a) Compute the projection of x onto y .
 - (b) Compute the residual vector.
 - (c) Verify numerically that the residual is orthogonal to y .
7. [E] Generate random 2×2 matrices.
 - (a) Compute their determinants using `np.linalg.det`.
 - (b) Use an `if` condition to identify nearly singular matrices.
8. [M] For an invertible matrix A and vector b :
 - (a) Solve $Ax = b$ using `np.linalg.solve`.
 - (b) Solve the same system using $A^{-1}b$.
 - (c) Compare the two solutions numerically.
9. [M] Let $x, y \in \mathbb{R}^n$ and consider the model $y \approx wx$.
 - (a) Compute the error $\|y - wx\|_2^2$ for different values of w .
 - (b) Find the value of w that minimizes the error.
 - (c) Plot the error as a function of w .
10. [C] Write a program that:
 - generates a random dataset,
 - normalizes each vector if needed,
 - computes pairwise cosine similarities,
 - prints a warning if highly similar vectors are detected.

::: End of Assignment Sheet :::